THE WEST AFRICAN EXAMINATIONS COUNCIL

BECE 2026

MATHEMATICS 2

FINAL MARKING SCHEME

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THE WEST AFRICAN EXAMINATIONS COUNCIL, ACCRA

BASIC EDUCATION CERTIFICATE EXAMINATION FOR SCHOOL CANDIDATES, 2026

FINAL MARKING SCHEME

MATHEMATICS 2

QUESTION NO.	SOLUTION	MARKS
1. (a) (i)	Prime Numbers = $\{2, 3, 5, 7, 11\}$ any three	M1 for any three correct
	for all correct	A1 for all correct
(ii)	Numbers not prime = $\{1, 4, 6, 8, 9, 10\}$	B1 for listing non-primes or 11-5=6 or listing of K
	$= 6$ or	B1 for 11
	Probability (Number not prime) $= \frac{6}{11}$	M1 A1
(b)	$(x + y)(2m - n) - m(x + y)$	M1 for factorising $(x + y)$ out
	$(x + y)(2m - n - m)$ at least one term correct	M1 for simplifying (at least one term correct)
	$(x + y)(m - n)$	A1 for $(x + y)(m - n)$
(c)	8 sheets = GH¢68.00	
	$= 35 - 8$ insist on 35-8 to get M1	M1 for subtracting
	$= 27$	A1 for 27
	= GH¢11.00 × 27	M1 for multiplying by 11
	= GH¢297.00	A1 for GH¢297.00
	Amount earned = GH¢68.00 + GH¢297	M1 for adding
	= GH¢365.00	A1 for GH¢365.00
	-1ou/wu or 2dp once only	[15 marks]

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QUESTION NO.	SOLUTION	MARKS
2. (a) (i)	$r + \begin{pmatrix}2\\5\end{pmatrix} = \begin{pmatrix}-3\\8\end{pmatrix}$	M1 for correct equation or equivalent
	$r = \begin{pmatrix}-3\\8\end{pmatrix} - \begin{pmatrix}2\\5\end{pmatrix}$	M1 for solving
	$r = \begin{pmatrix}-5\\3\end{pmatrix}$	A1 for $\begin{pmatrix}-5\\3\end{pmatrix}$
(ii)	Image = r + point	
	$Q^1 = \begin{pmatrix}-5\\3\end{pmatrix} + \begin{pmatrix}-4\\-6\end{pmatrix}$	M1 for correct equation or equivalent
	$Q^1 = \begin{pmatrix}-9\\-3\end{pmatrix}$	A1 for $\begin{pmatrix}-9\\-3\end{pmatrix}$
	$Q^1\ (-9, -3)$ Points must be in coordinate	A1 for $Q^1(-9, -3)$
(b)		
	Amount paid to Richard = $\frac{1}{3} \times$ GH¢9,000.00	M1 for dividing by 3
	= GH¢3,000.00	A1 for GH¢3,000.00
	Amount left = GH¢9000 – GH¢3,000	
	= GH¢6000.00	A1 for GH¢6,000.00
	Ratio Joyce : Richard	
	10 : 5	
	2 : 1	
	Total parts = 3	B1 for 3 or equivlent
	Joyce = $\frac{2}{3} \times$ GH¢6000	M1 for $\frac{2}{3} \times$ GH¢6000
	= GH¢4,000.00	A1 for GH¢4,000.00
	Richard = $\frac{1}{3} \times$ GH¢6000	
	= GH¢2,000.00	A1 for GH¢2,000.00
	Amount received by Richard	
	= GH¢2,000.00 + GH¢3,000.00	M1 for adding
	= GH¢5,000.00	A1 for GH¢5,000.00
	-1 ou/wu or 2dp once only	[15 Ma

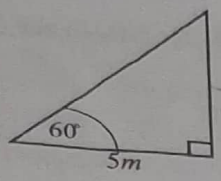
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QUESTION NO.	SOLUTION	MARKS
3. (a)	$\frac{10}{100} \times$ GH¢25.00 = GH¢2.50	M1A1
	Asana's new wage = GH¢25 + 2.50	M1
	= GH¢27.50	A1
	Asana's amount for 30 days = 27.50 × 30	M1 for multiplying by 30
	Asana's new pay = GH¢825.00	A1 for GH¢825.00
(b) (i)	Perimeter = $5m + 8m + 3m + 5m + (5m - 3m) + (8m - 5m)$	M1 for $(5m - 3m)$ or $(8m - 5m)$
	$= 26m$	M1 adding all sides
		A1 for $26m$
(ii)	Area of the floor = $(8 \times 3)m^2 + (3 \times 2)m^2$	M1 for $(8 \times 3)m^2$ or $(5 \times 3)m^2$
	OR $(5\times3)m^2 + (5\times3)m^2$	M1 for $(3 \times 2)m^2$ or $(5 \times 3)m^2$
	$= 24m^2 + 6m^2$ or $15m^2 + 15m^2$	M1 for adding
	$= 30m^2$	A1 for $30m^2$
(iii)	Cost of carpeting the floor	
	$= 30 \times 20$	M1 for multiplying
	= GH¢600.00	A1 for GH¢600.00
	-1ou/wu or 2dp once only	[15 marks]

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QUESTION NO.	SOLUTION	MARKS
4. (a) (i)	Aku's yearly income = 12 × 5000	
	= GH¢60,000.00	B1 for GH¢60,000.00
(ii)	Brako's yearly income = 12 × GH¢6500	M1
	= GH¢6500 (10 + 2)	
	= GH¢65000 + GH¢13000	
	= GH¢78,000.00	A1 with evidence
	Dagadu's yearly income = GH¢4200 × 12	M1
	= 4200 × (10 + 2)	
	= 42,000 + 8,400	
	= GH¢50,400.00	A1 with evidence
	Income difference = GH¢60,000 – GH¢50,400	
	= GH¢9,600.00	B1 for GH¢9,600.00
(iii)	Total yearly income = GH¢60,000.00	
	+ GH¢78,000.00	M1 for adding candidate's income
	GH¢50,400.00	
	GH¢188,400.00	A1 for GH¢188,400.00

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QUESTION NO.	SOLUTION	MARKS
4. (b) (i)	 for diagram (-½ ee). if No diagram everything is wrong	B1 for diagram $\frac{-1}{2}$ ee (60°, 5m)
(ii) (α)	Length of ladder $= \frac{5}{\cos 60°}$ $\cos 60 = \frac{5}{\text{Length of ladder}}$ (other alternative) $= \frac{5}{1/2}$ $= 10.0m$	M1 for correct trigonometry ratio M1 for simplifying A1 for $10.0m$
(β)	$h = 5 \tan 60°$ $\tan 60 = \frac{h}{5}$ (other formula) $= 5 \times 1.732$ $= 8.66$ $= 8.7m$ No evidence is needed.	M1 for correct trigonometry ratio M1 for multiplying A1 for $8.7m$
	-1ou/wu once only	[15 Marks]

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QUESTION NO.	SOLUTION	MARKS
5. (a)	$= \frac{6\sqrt{2}}{3\sqrt{2} - 2\sqrt{3}}$	M1 for reducing any surd to basic form
	$= \frac{6\sqrt{2}(3\sqrt{2}+2\sqrt{3})}{(3\sqrt{2}-2\sqrt{3})(3\sqrt{2}+2\sqrt{3})}$	M1 for rationalizing
	$= \frac{18(\sqrt{2})^2 + 12\sqrt{2}(\sqrt{3})}{18-12}$	M1 for any one term correct for numerator or denominator
	$= \frac{18(2)+12\sqrt{6}}{6}$	
	$= \frac{36}{6} + \frac{12\sqrt{6}}{6}$	M1 for dividing by 6
	$= 6 + 2\sqrt{6}$	A1 for $6 + 2\sqrt{6}$
(b)	$\frac{1}{2}(2x+1) \geq \frac{1}{3}x + 1\frac{9}{10}$	
	$\frac{1}{2}(2x+1) \geq \frac{1}{3}x + \frac{19}{10}$	B1 for $\frac{19}{10}$
	$15(2x+1) \geq 10x + 3(19)$	M1 for clearing fractions
	$30x + 15 \geq 10 + 57$	M1 for correct expansion
	$30x - 10x \geq 57 - 15$	M1 for solving
	$20x \geq 42$	
	$x \geq \frac{42}{20}$	
	$x \geq 2\frac{1}{10}$ accept $\frac{21}{10}$ or 2.1	A1 for $x \geq 2\frac{1}{10}$
(c)	Circular Track $= 2\pi r$	
	$= 2 \times \frac{22}{7} \times 70$	M1 for correct substitution
	$= 44 \times 10$	M1 for simplifying
	$= 440m$	A1 for $440m$
	$4\ times = 4(440)$	M1 for multiplying by 4
	$= 1760m$	A1 for $1760m$
		[15 Marks]

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QUESTION NO.	SOLUTION	MARKS
6. (a)		

Shoe size (x)	Tally	Frequency (f)	fx
4	𝍸 |	7	28
5	𝍸 𝍸 |	10	50
6	𝍸 𝍸 |||	8	48
7	𝍸 |||	5	35
		$\Sigma f = 30$	$\Sigma fx = 161$

B1 for Tally
M1 for any two frequency correct
A1 for all frequencies correct
M1 for any $f(x)$ correct
A1 for all correct
B1 for $\Sigma f = 30$
M1 adding fx's
A1 for 161

(b) (i) Shoe size 5 — **B1 for 5**

(ii) There are more, many, large learners with shoe size 5 — **B1 for any of the words (more, many or large) correct**

(iii) Shoe size 7 — **B1 for 7**

(iv) There are less or small number of learners with shoe size 7. — **B1 for any of the words (less, small) or equivalent**

(c) Mean $= \frac{161}{30}$ — **M1**

$= 5.367$ — **M1 for simplifying with evidence**

$= 5$ — **A1 for 5**

[15 Marks]

1. 15/05/2026
2. 15/5/26
3. 15/5/2026
4. 15/5/2026